

Pre-Lab 1 Generation of Digital Signals and Signal Spectral Analysis Using MATLAB

OBJECTIVES

1. To learn how to generate typical digital signals such as white noise with the Gaussian distribution and sum of sinusoids.
2. To investigate bandwidths of the generated digital signals and speech waveform.

PROCEDURE:

1. Generation of white noise with the Gaussian distribution

Generate the white noise with the Gaussian distribution using the following code:

```
x = 5* randn([1,4096]); % generate 4096 noise samples
```

- a. Make a plot of the white noise.

Plot the first 50 samples (discrete-time signal style) using the Matlab function “stem()”.

Hint: MATLAB>>Figure; subplot(2,1,1);stem(x(1:50));

- b. Use fft() to compute DFT coefficients, and plot and examine the magnitude spectrum of the signal $x(n)$ assuming the sampling rate is 8000 Hz. To obtain frequency components:

```
MATLAB>> Ak=abs(fft(x))/length(x);
```

To plot the spectrum:

```
MATLAB>> fs=8000;
```

```
MATLAB>>k=0:1:length(x)-1; %frequency index
```

```
MATLAB> f=k*fs/length(x); % convert it to Hz
```

```
MATLAB>subplot(2,1,2); plot(f,Ak);
```

Label and print your graph.

What frequency range should you read? _____

What is the frequency resolution? _____

- c. Listen to the generated signal using the sampling rate of 8000 Hz.

```
MATLAB>sound(x/max(abs(x)),8000);
```

2. Generation of the sum of digital sinusoids

Generate three sinusoids with the following amplitudes and phases

$$x_1(t) = 5 \cos[2\pi(500)t] ; \quad x_2(t) = 5 \cos[2\pi(1200)t + 0.25\pi] ; \quad x_3(t) = 5 \cos[2\pi(1800)t + 0.5\pi]$$

- a. Make a plot of all three signals over a range of t that will exhibit approximately 0.01 second. Each signal will be sampled using a sampling rate of 8000 Hz, a step size (sampling period) for time t is $1/8000=0.000125$ second

Hint: MATLAB>> t=0:(1/8000):0.5; figure; subplot(3,1,1);plot(t,x1); axis([0 0.01 -6 6]);

Verify that the phases of all three signals are correct, and verify that each one has the correct maximum amplitude.

Calculated $x_1(0) =$	$x_{1\max} =$	Measured $x_1(0) =$	$x_{1\max} =$
Calculated $x_2(0) =$	$x_{2\max} =$	Measured $x_2(0) =$	$x_{2\max} =$
Calculated $x_3(0) =$	$x_{3\max} =$	Measured $x_3(0) =$	$x_{3\max} =$

- b. Create a program to generate the sum of sinusoids by adding three generated sinusoids

$$x(t) = x_1(t) + x_2(t) + x_3(t) .$$

Make a plot of $x(t)$ over the same range of time as used in (1).

MATLAB>>figure; subplot(2,1,1);plot(t,x);axis([0 0.01 -6 6]);

What is the sampling rate? _____

- c. Use fft() to compute DFT coefficients, plot and examine the spectrum of the signal $x(n)$ (frequency components).

To obtain frequency components:

MATLAB> Ak=abs(fft(x))/length(x);

To plot the spectrum:

MATLAB>> fs=8000;

MATLAB>>k=0:length(x)-1; %k is the frequency bin (frequency index)

MATLAB>> f=k*fs/length(x); % convert xx to Hz

MATLAB>>subplot(2,1,2);plot(f,Ak);

Label and print your graph.

What frequencies are shown in the signal spectral plot? _____

What is the maximum frequency value that you can see? _____

What is the frequency resolution based on the MATLAB program? _____

- d. Listen to and compare the generated signals

MATLAB>>sound(x1/max(abs(x1)), 8000);pause

MATLAB>>sound(x2/max(abs(x2)), 8000);pause

MATLAB>>sound(x3/max(abs(x3)), 8000);pause

MATLAB>>sound(x/max(abs(x)), 8000);

3. Generation of the multiplication signal of digital sinusoids

Generate two sinusoids with the following amplitudes and phases

$$x_1(t) = 5 \cos[2\pi(500)t] ; \quad x_2(t) = 5 \cos[2\pi(1200)t + 0.25\pi]$$

- a. Make a plot of all two signals over a range of t that will exhibit approximately 0.01 second. Each signal will be sampled using the sampling rate of 8000 Hz, the step size (sampling period) for time t is $1/8000=0.000125$ second

Hint: MATLAB>> t=0:(1/8000):0.5; figure; subplot(4,1,1);plot(t,x1); axis([0 0.01 -6 6]);

Use MATLAB routines “max()” and “mean()” to determine the maximum value and root-mean squared value (hint: sqrt(mean(x1.*x1)))

$$x_1 \text{ max} = \quad \quad \quad x_{1rms} =$$

$$x_2 \text{ max} = \quad \quad \quad x_{2rms} =$$

- b. Create a program to generate the multiplication of sinusoids by using: $y(t) = x_1(t) \times x_2(t)$. Hint:

MATLAB>>y=x1.*x2;

Make a plot of $y(t)$ over same range of time as used in (1).

MATLAB>>figure; subplot(4,1,3);plot(t,y);axis([0 0.01 -6 6]);

- c. Use `fft()` to compute DFT coefficients, plot and examine the spectrum of the signal $y(n)$ (frequency components).

To obtain frequency components:

```
MATLAB>> Ak=abs(fft(y))/length(y);
```

To plot the spectrum:

```
MATLAB>> fs=8000;
```

```
MATLAB>> k=0:length(x)-1; %k is the frequency bin (frequency index)
```

```
MATLAB>> f=k*fs/length(y); % convert xx to Hz
```

```
MATLAB>> subplot(4,1,4);plot(f,Ak);
```

Label and print your graph.

What frequencies having the amplitude spectral peaks are shown in the signal spectral plot?

What is the maximum amplitude for each amplitude spectral peak ? _____

What is the frequency resolution based on the MATLAB program? _____

- d. Listen to and compare the generated signals

```
MATLAB>> pause
```

```
MATLAB>> sound(x1/max(abs(x1)), 8000);pause
```

```
MATLAB>> sound(x2/max(abs(x2)), 8000);pause
```

```
MATLAB>> sound(y/max(abs(y)), 8000);
```

4. Digital speech waveform

- a. Plot the speech wave sampled at 8000 Hz provided by your instructor:

```
MATLAB>load we.dat; %get speech data
```

```
MATLAB>>x=we;
```

Plot the speech (discrete-time signals)

Hint: MATLAB>>figure;

```
MATLAB>> n=0:length(x)-1; % recover time index
```

```
MATLAB>> T=1/8000;
```

```
MATLAB>> t=n*T; % recover time in sec.
```

```
MATLAB>> subplot(2,1,1);plot(t, x);
```

- b. Use `fft()` to compute DFT coefficients, plot and examine the spectrum of the speech signal $x(n)$. To obtain frequency components:

```
MATLAB>> Ak=abs(fft(x))/length(x);
```

To plot the spectrum:

```
MATLAB>> k=0:length(x)-1; %frequency bin (frequency index)
```

```
MATLAB>> f=k*8000/length(Ak); % convert xx to Hz
```

```
MATLAB>> subplot(2,1,2);plot(f,Ak);
```

Label and print your graph.

What is the maximum frequency (bandwidth) that you can approximately see? _____

Note: the sampling rate for the speech data is 8000 Hz.

What is the frequency resolution based on the MATALB program? _____

c. Listen to the speech signal

MATALB>>sound(x/max(abs(x)),8000);

Instructor verification_____

5. Organize and submit your plots and program lists